MIPT & YSDA · AUTUMN 2026

Deep Generative Models

Lecture 6

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Recap of Previous Lecture

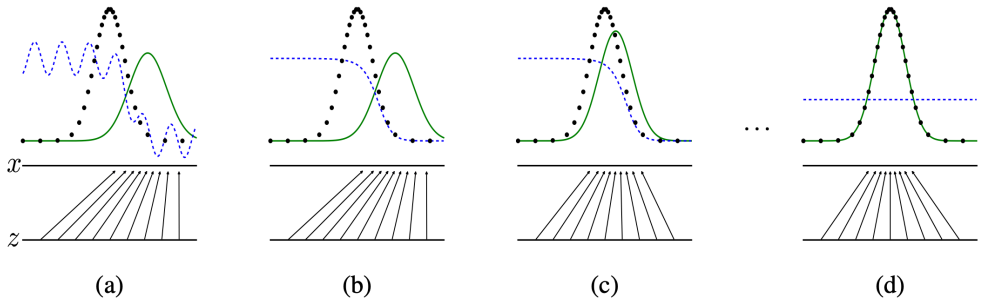
Objective

$$\min_{\boldsymbol{\theta}} \max_{\boldsymbol{\phi}} \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D_{\boldsymbol{\phi}}(\mathbf{x}) + \mathbb{E}_{p(\mathbf{z})} \log(1 - D_{\boldsymbol{\phi}}(\mathbf{G}_{\boldsymbol{\theta}}(\mathbf{z}))) \right]$$

Optimal Discriminator and Jensen-Shannon Divergence

For fixed G, write the bracketed objective as $V(G, D)$:

$$D^*(\mathbf{x}) = \frac{p_{\text{data}}(\mathbf{x})}{p_{\text{data}}(\mathbf{x}) + p_{\boldsymbol{\theta}}(\mathbf{x})}, \qquad V(G, D^*) = 2\, D_{\text{JS}}(p_{\text{data}}(\mathbf{x}) \,\|\, p_{\boldsymbol{\theta}}(\mathbf{x})) - 2 \log 2.$$



Mode collapse refers to the phenomenon where the generator in a GAN produces only one or a few different modes of the distribution.

Goodfellow I. J. et al. Generative Adversarial Networks, 2014
Metz L. et al. Unrolled Generative Adversarial Networks, 2016

Recap of Previous Lecture

GAN losses also train the reconstruction stage of a **latent generative model**.

Stage 1: Train the Autoencoder

A fixed pretrained network F defines the perceptual loss for reconstruction $\hat{\mathbf{x}}$:

$$\mathcal{L}_{\text{perc}} = \|F(\mathbf{x}) - F(\hat{\mathbf{x}})\|_2^2.$$

Train a discriminator D to separate real and reconstructed image patches; train the autoencoder to fool D.

A KL-autoencoder also uses weak KL regularization.

Stage 2: Freeze the Autoencoder and Learn a Prior

Fit the distribution of its latent codes — the **aggregated posterior**.

VQGAN: sample a code map $\mathbf{c}$ from an AR prior, look up its embeddings $\mathbf{z}_q(\mathbf{c})$, then decode.

$$\mathbf{c} \sim p_{\psi}(\mathbf{c}) \;\longrightarrow\; \mathbf{z}_q(\mathbf{c}) \;\longrightarrow\; \hat{\mathbf{x}}.$$

KL-autoencoder: a diffusion prior over continuous latents.

$$\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \;\longrightarrow\; \mathbf{z} \sim p_{\psi}(\mathbf{z}) \;\longrightarrow\; \hat{\mathbf{x}}.$$

Esser P., Rombach R., Ommer B. Taming Transformers for High-Resolution Image Synthesis, 2021
Rombach R. et al. High-Resolution Image Synthesis with Latent Diffusion Models, 2022

Recap of Previous Lecture

Standard GAN Objective

$$\min_{\boldsymbol{\theta}} \max_{\boldsymbol{\phi}} \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D_{\boldsymbol{\phi}}(\mathbf{x}) + \mathbb{E}_{p(\mathbf{z})} \log(1 - D_{\boldsymbol{\phi}}(\mathbf{G}_{\boldsymbol{\theta}}(\mathbf{z}))) \right]$$

Theoretical Results

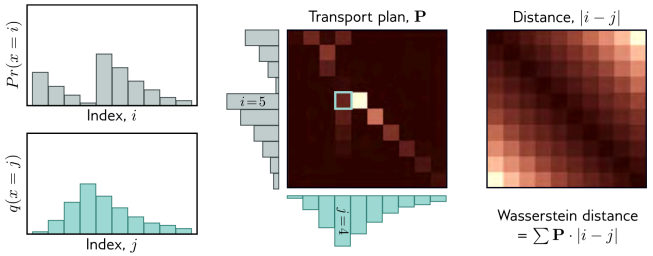
- Both the data distribution $p_{\text{data}}(\mathbf{x})$ and the generative distribution $p_{\boldsymbol{\theta}}(\mathbf{x})$ are supported on low-dimensional manifolds.
- If $p_{\text{data}}(\mathbf{x})$ and $p_{\boldsymbol{\theta}}(\mathbf{x})$ are disjoint, a smooth optimal discriminator can exist!

For such low-dimensional, disjoint manifolds:

$$D_{\text{KL}}(p_{\text{data}} \,\|\, p_{\boldsymbol{\theta}}) = D_{\text{KL}}(p_{\boldsymbol{\theta}} \,\|\, p_{\text{data}}) = \infty, \qquad D_{\text{JS}}(p_{\text{data}} \,\|\, p_{\boldsymbol{\theta}}) = \log 2$$

Weng L. From GAN to WGAN, 2019
Arjovsky M., Bottou L. Towards Principled Methods for Training Generative Adversarial Networks, 2017

Recap of Previous Lecture



Wasserstein Distance

$$W(\pi\|p) = \inf_{\gamma\in\Gamma(\pi,p)} \mathbb{E}_{(\mathbf{x}_1,\mathbf{x}_2)\sim\gamma}\|\mathbf{x}_1 - \mathbf{x}_2\|$$

$$= \inf_{\gamma\in\Gamma(\pi,p)} \int \|\mathbf{x}_1 - \mathbf{x}_2\|\gamma(\mathbf{x}_1,\mathbf{x}_2)d\mathbf{x}_1 d\mathbf{x}_2$$

- $\gamma(\mathbf{x}_1, \mathbf{x}_2)$ is the transport plan: the amount of "dirt" assigned from $\mathbf{x}_1$ to $\mathbf{x}_2$.
- $\Gamma(\pi, p)$ denotes the set of all joint distributions $\gamma(\mathbf{x}_1, \mathbf{x}_2)$ with marginals π and p ($\int \gamma(\mathbf{x}_1, \mathbf{x}_2)d\mathbf{x}_1 = p(\mathbf{x}_2)$, $\int \gamma(\mathbf{x}_1, \mathbf{x}_2)d\mathbf{x}_2 = \pi(\mathbf{x}_1)$).
- $\gamma(\mathbf{x}_1, \mathbf{x}_2)$ is the mass, $\|\mathbf{x}_1 - \mathbf{x}_2\|$ is the distance.

Simon J.D. Prince. Understanding Deep Learning, 2023
Arjovsky M., Chintala S., Bottou L. Wasserstein GAN, 2017

Recap of Previous Lecture

On a compact data space $\mathcal{X}$:

Dependence on Generator Parameters

For a feedforward generator $\mathbf{G}_{\boldsymbol{\theta}}$ with Lipschitz activations and $\mathbb{E}_{p(\mathbf{z})}\|\mathbf{z}\| < \infty$, $W(p_{\text{data}}\|p_{\boldsymbol{\theta}})$ is continuous in $\boldsymbol{\theta}$ and differentiable almost everywhere.

Convergence of Distributions

For a fixed distribution π and a sequence p_t on $\mathcal{X}$, as $t \to \infty$:

$$D_{\text{KL}}(\pi\|p_t) \to 0 \quad \Longrightarrow \quad D_{\text{JS}}(\pi\|p_t) \to 0 \quad \Longrightarrow \quad W(\pi\|p_t) \to 0$$

The first implication also holds for $D_{\text{KL}}(p_t\|\pi) \to 0$; the converses need not hold.

> **Wasserstein can capture distributions getting closer even when their supports do not overlap.**

Recap of Previous Lecture

Theorem (Kantorovich-Rubinstein Duality)

$$W(p_{\text{data}} \| p_{\boldsymbol{\theta}}) = \frac{1}{K} \max_{\|f\|_L \leq K} \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} f(\mathbf{x}) - \mathbb{E}_{p_{\boldsymbol{\theta}}(\mathbf{x})} f(\mathbf{x}) \right],$$

where $f : \mathbb{R}^m \to \mathbb{R}$ is K-Lipschitz continuous ($\|f\|_L \leq K$).

WGAN Objective

$$\min_{\boldsymbol{\theta}} W(p_{\text{data}} \| p_{\boldsymbol{\theta}}) \approx \min_{\boldsymbol{\theta}} \max_{\boldsymbol{\phi} \in \boldsymbol{\Phi}} \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} f_{\boldsymbol{\phi}}(\mathbf{x}) - \mathbb{E}_{p(\mathbf{z})} f_{\boldsymbol{\phi}}(\mathbf{G}_{\boldsymbol{\theta}}(\mathbf{z})) \right].$$

- The discriminator D is replaced by function f: in WGAN, it is known as the **critic**, which is *not* a classifier.
- If the weights $\boldsymbol{\phi}$ are restricted to a compact set $\boldsymbol{\Phi} = [-c, c]^d$, then $f_{\boldsymbol{\phi}}(\mathbf{x})$ is K-Lipschitz continuous.

$$\begin{aligned} K \cdot W(p_{\text{data}} \| p_{\boldsymbol{\theta}}) &= \max_{\|f\|_L \leq K} \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} f(\mathbf{x}) - \mathbb{E}_{p_{\boldsymbol{\theta}}(\mathbf{x})} f(\mathbf{x}) \right] \\ &\geq \max_{\boldsymbol{\phi} \in \boldsymbol{\Phi}} \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} f_{\boldsymbol{\phi}}(\mathbf{x}) - \mathbb{E}_{p_{\boldsymbol{\theta}}(\mathbf{x})} f_{\boldsymbol{\phi}}(\mathbf{x}) \right] \end{aligned}$$

Recap of Previous Lecture

Frechet Inception Distance (FID)

Use Inception image embeddings $\mathbf{z} = \mathbf{f}(\mathbf{x})$; compare Gaussian approximations of their distributions:

$$\begin{aligned}\mathrm{FID}(p_{\text{data}}, p_{\boldsymbol{\theta}}) = \|\boldsymbol{\mu}_{\text{data}} - \boldsymbol{\mu}_{\boldsymbol{\theta}}\|^2 \\ + \mathrm{tr}\left[\boldsymbol{\Sigma}_{\text{data}} + \boldsymbol{\Sigma}_{\boldsymbol{\theta}} - 2\left(\boldsymbol{\Sigma}_{\text{data}}^{1/2}\boldsymbol{\Sigma}_{\boldsymbol{\theta}}\boldsymbol{\Sigma}_{\text{data}}^{1/2}\right)^{1/2}\right].\end{aligned}$$

FID depends on the feature network, Gaussian approximation and sample size; it may not correlate with human evaluation.

CMMD: Comparing CLIP Image Embeddings

For CLIP image-embedding distributions P, Q, use a Gaussian kernel k:

$$\mathrm{MMD}_k^2(P, Q) = \mathbb{E}k(\mathbf{z}, \mathbf{z}') + \mathbb{E}k(\mathbf{w}, \mathbf{w}') - 2\mathbb{E}k(\mathbf{z}, \mathbf{w}).$$

Here $\mathbf{z}, \mathbf{z}' \sim P$ and $\mathbf{w}, \mathbf{w}' \sim Q$ are mutually independent. This compares more than means and covariances.

Heusel M. et al. GANs Trained by a Two Time-Scale Update Rule Converge to a Local Nash Equilibrium, 2017
Jayasumana S. et al. Rethinking FID: Towards a Better Evaluation Metric for Image Generation, 2024

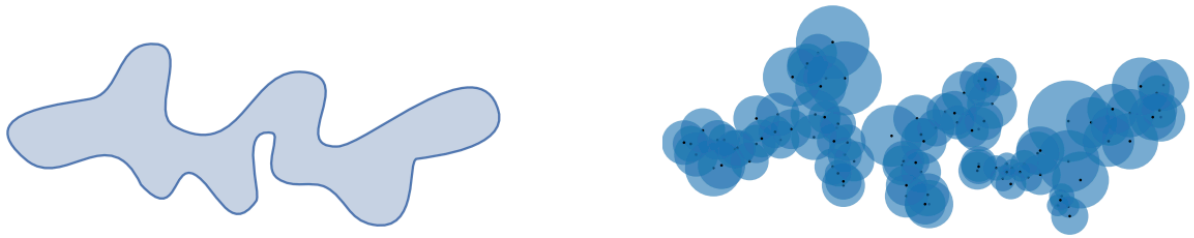
Recap of Previous Lecture

- $\mathcal{S}_{\text{data}} = \{\mathbf{x}_i\}_{i=1}^n \sim p_{\text{data}}(\mathbf{x})$ - real samples;
- $\mathcal{S}_{\boldsymbol{\theta}} = \{\mathbf{x}_i\}_{i=1}^n \sim p_{\boldsymbol{\theta}}(\mathbf{x})$ - generated samples.

Define a binary function:

$$\mathbb{I}(\mathbf{x}, \mathcal{S}) = \begin{cases} 1, & \text{if } \exists\, \mathbf{x}' \in \mathcal{S} : \|\mathbf{x} - \mathbf{x}'\|_2 \leq \|\mathbf{x}' - \text{NN}_k(\mathbf{x}', \mathcal{S})\|_2; \\ 0, & \text{otherwise.} \end{cases}$$

$$\text{Pr}(\mathcal{S}_{\text{data}}, \mathcal{S}_{\boldsymbol{\theta}}) = \frac{1}{n} \sum_{\mathbf{x} \in \mathcal{S}_{\boldsymbol{\theta}}} \mathbb{I}(\mathbf{x}, \mathcal{S}_{\text{data}}); \quad \text{Rec}(\mathcal{S}_{\text{data}}, \mathcal{S}_{\boldsymbol{\theta}}) = \frac{1}{n} \sum_{\mathbf{x} \in \mathcal{S}_{\text{data}}} \mathbb{I}(\mathbf{x}, \mathcal{S}_{\boldsymbol{\theta}}).$$



(a) True manifold (b) Approx. manifold

Embed the samples using a pretrained network (as in FID).

Recap of Previous Lecture

CLIP: Learn from Image–Caption Pairs

$$s_{\text{CLIP}}(\mathbf{x}, t) = \cos\left(\mathbf{f}_{\text{img}}(\mathbf{x}), \mathbf{f}_{\text{text}}(t)\right).$$

PickScore learns human preferences; **VQAScore / GenEval / DPG-Bench** assess compositional correctness beyond CLIP similarity.

Human Evaluation

No automated metric is perfect. Validate model quality with human judgments across multiple criteria. For image editing, assess:

1. **Instruction relevance:** Was the requested change made?
2. **Preservation:** Did content that should stay unchanged remain intact?
3. **Visual defects:** Were artifacts or distortions introduced?

Radford A. et al. Learning Transferable Visual Models From Natural Language Supervision, 2021 · Kirstain Y. et al. Pick-a-Pic, 2023
Lin Z. et al. Evaluating Text-to-Visual Generation with Image-to-Text Generation, 2024
Ghosh D. et al. GenEval, 2023 · Hu X. et al. ELLA (DPG-Bench), 2024 · Yandex. Alice AI ART 2.0

Outline

01 Langevin Dynamics

02 Score Matching

Denoising Score Matching
Noise-Conditioned Score Network (NCSN)

03 Forward Gaussian Diffusion Process

Outline

01 **Langevin Dynamics**

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Energy-Based Models

Unnormalized Density

$$p_{\boldsymbol{\theta}}(\mathbf{x}) = \frac{\hat{p}_{\boldsymbol{\theta}}(\mathbf{x})}{Z_{\boldsymbol{\theta}}}, \qquad \text{where } Z_{\boldsymbol{\theta}} = \int \hat{p}_{\boldsymbol{\theta}}(\mathbf{x}) d\mathbf{x}$$

- $\hat{p}_{\boldsymbol{\theta}}(\mathbf{x})$ can be any non-negative function.
- If we reparameterize as $\hat{p}_{\boldsymbol{\theta}}(\mathbf{x}) = \exp(-f_{\boldsymbol{\theta}}(\mathbf{x}))$, we eliminate the non-negativity constraint.

Log-Density Gradient

The gradient of the normalized log-density equals that of the unnormalized log-density:

$$\nabla_{\mathbf{x}} \log p_{\boldsymbol{\theta}}(\mathbf{x}) = \nabla_{\mathbf{x}} \log \hat{p}_{\boldsymbol{\theta}}(\mathbf{x}) - \nabla_{\mathbf{x}} \log Z_{\boldsymbol{\theta}} = \nabla_{\mathbf{x}} \log \hat{p}_{\boldsymbol{\theta}}(\mathbf{x})$$

- Suppose we already have this density (normalized or not) $p_{\boldsymbol{\theta}}(\mathbf{x})$.
- How can we sample from the model?

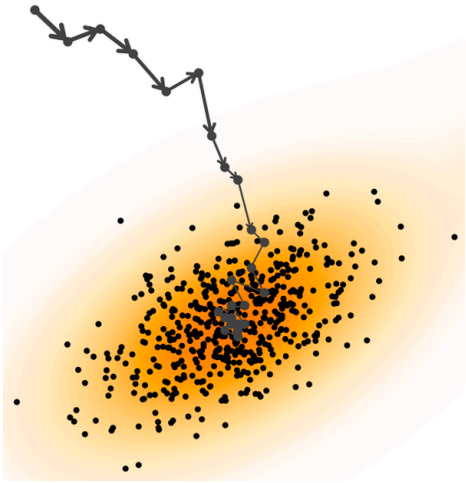
Langevin Dynamics

Theorem (Informal)

Let $\mathbf{x}_0$ be a random vector. Under mild regularity conditions, samples from the following dynamics will eventually follow $p_{\boldsymbol{\theta}}(\mathbf{x})$ (for sufficiently small η and large l):

$$\mathbf{x}_{l+1} = \mathbf{x}_l + \frac{\eta}{2} \cdot \nabla_{\mathbf{x}_l} \log p_{\boldsymbol{\theta}}(\mathbf{x}_l) + \sqrt{\eta} \cdot \boldsymbol{\epsilon}_l, \qquad \boldsymbol{\epsilon}_l \sim \mathcal{N}(0, \mathbf{I}).$$

- What if $\boldsymbol{\epsilon}_l = \mathbf{0}$?
- The density $p_{\boldsymbol{\theta}}(\mathbf{x})$ is the **stationary** distribution of the Markov chain.
- The gradient is taken with respect to $\mathbf{x}$, not $\boldsymbol{\theta}$.
- $\nabla_{\mathbf{x}} \log p_{\boldsymbol{\theta}}(\mathbf{x})$ defines a vector field.



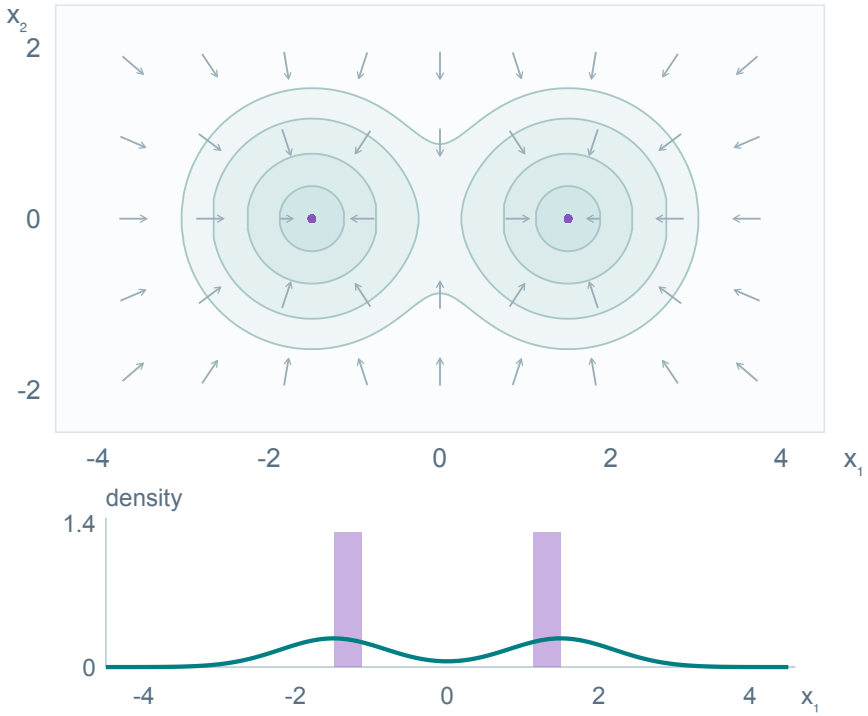
Knowing $p_{\boldsymbol{\theta}}(\mathbf{x})$ up to a normalizing constant lets us use Langevin dynamics to obtain approximate samples $\mathbf{x}_l \sim p_{\boldsymbol{\theta}}$ for large l.

Lai C. H. et al. The principles of diffusion models, 2025.

Langevin Dynamics: Why Add Noise?

$$p_{\theta}(\mathbf{x}) = \tfrac{1}{2}\mathcal{N}(\boldsymbol{\mu}_{-}, 0.7^2\mathbf{I}) + \tfrac{1}{2}\mathcal{N}(\boldsymbol{\mu}_{+}, 0.7^2\mathbf{I}), \qquad \boldsymbol{\mu}_{\pm} = (\pm 1.5, 0), \quad \eta = 0.04.$$

Noise off: approach the modes Step 300



Density axis adapts to the histogram.

Noise on: explore the density Step 300



Density axis adapts to the histogram.

Finite step size and finite simulation time give approximate sampling.

Outline

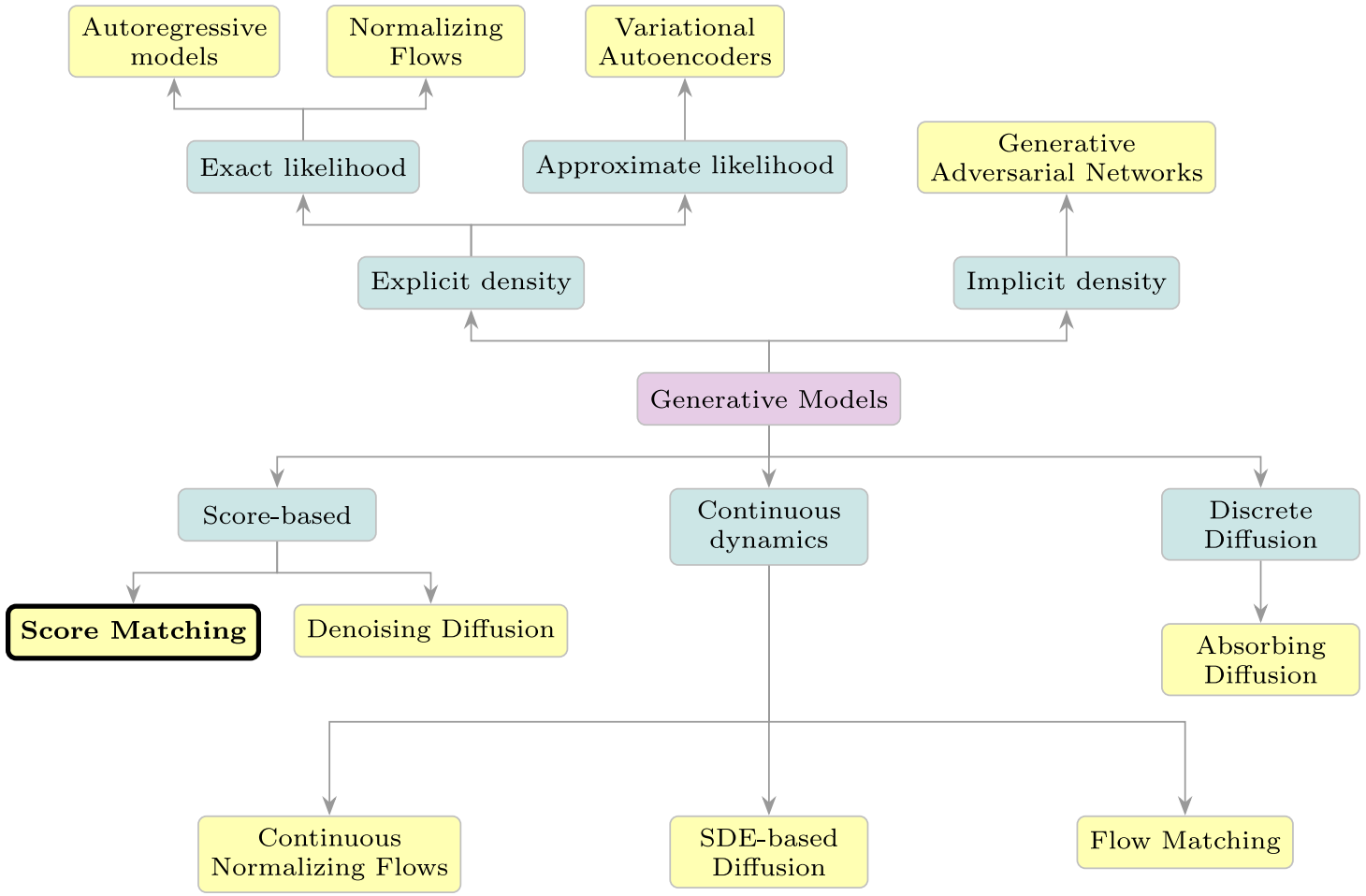
01 Langevin Dynamics

02 Score Matching

Denoising Score Matching
Noise-Conditioned Score Network (NCSN)

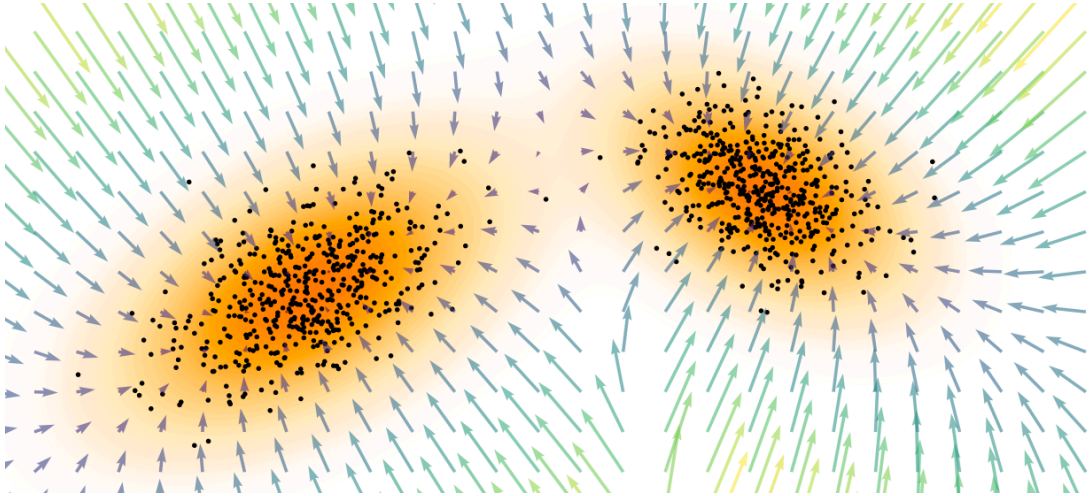
03 Forward Gaussian Diffusion Process

Generative Models Taxonomy



Score Function

$$\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}) = \nabla_{\mathbf{x}} \log p_{\boldsymbol{\theta}}(\mathbf{x})$$

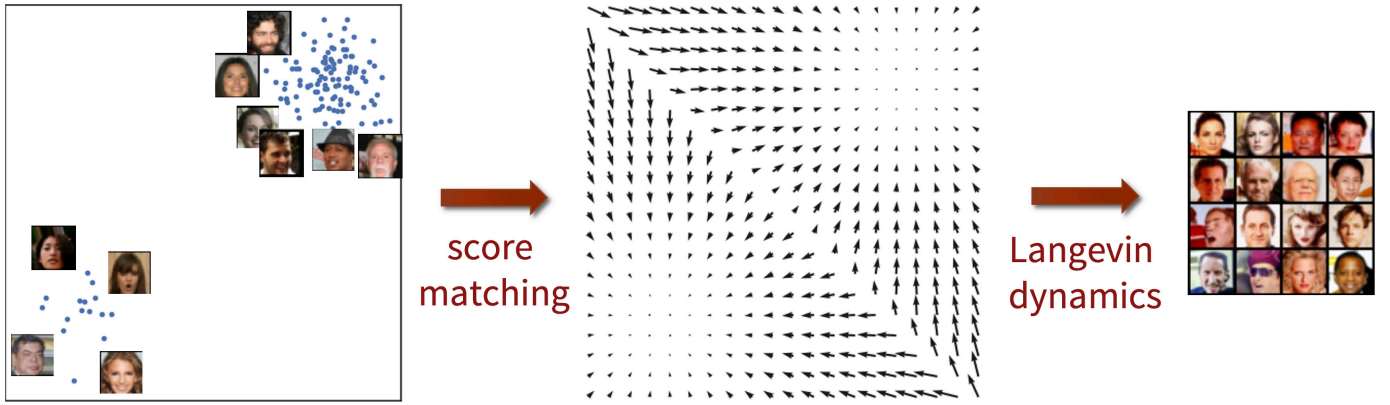


Langevin Dynamics

$$\begin{aligned}
\mathbf{x}_{l+1} &= \mathbf{x}_l + \frac{\eta}{2} \cdot \nabla_{\mathbf{x}_l} \log p_{\text{data}}(\mathbf{x}_l) + \sqrt{\eta} \cdot \boldsymbol{\epsilon}_l \\
&\approx \mathbf{x}_l + \frac{\eta}{2} \cdot \nabla_{\mathbf{x}_l} \log p_{\boldsymbol{\theta}}(\mathbf{x}_l) + \sqrt{\eta} \cdot \boldsymbol{\epsilon}_l \\
&= \mathbf{x}_l + \frac{\eta}{2} \cdot \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}_l) + \sqrt{\eta} \cdot \boldsymbol{\epsilon}_l.
\end{aligned}$$

Lai C. H. et al. The principles of diffusion models, 2025.

Score Matching



$$\{\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_N\} \overset{\text{i.i.d.}}{\sim} p(\mathbf{x}) \qquad \mathbf{s}_\theta(\mathbf{x}) \approx \nabla_{\mathbf{x}} \log p(\mathbf{x})$$

Fisher Divergence

$$\begin{aligned} D_F(p_{\text{data}}, p_{\boldsymbol{\theta}}) &= \frac{1}{2}\mathbb{E}_{p_{\text{data}}} \left\| \nabla_{\mathbf{x}} \log p_{\boldsymbol{\theta}}(\mathbf{x}) - \nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x}) \right\|_2^2 \\ &= \frac{1}{2}\mathbb{E}_{p_{\text{data}}} \left\| \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}) - \nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x}) \right\|_2^2 \to \min_{\boldsymbol{\theta}} \end{aligned}$$

Problem: We don't know $\nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x})$.

Why Is the Score Function Better than the Density?

Freedom from Normalization Constants

Many distributions are defined only up to an unnormalized density $\hat{p}(\mathbf{x})$:

$$\nabla_{\mathbf{x}} \log p(\mathbf{x}) = \nabla_{\mathbf{x}} \log \hat{p}(\mathbf{x}) - \underbrace{\nabla_{\mathbf{x}} \log Z}_{=0} = \nabla_{\mathbf{x}} \log \hat{p}(\mathbf{x})$$

The score function bypasses the intractable constant Z entirely!

A Complete Representation

The score $\mathbf{s}(\mathbf{x}) = \nabla_{\mathbf{x}} \log p(\mathbf{x})$ determines $\log p(\mathbf{x})$ up to the additive constant $\log p(\mathbf{x}_0)$:

$$\log p(\mathbf{x}) = \log p(\mathbf{x}_0) + \int_0^1 \mathbf{s}(\mathbf{x}_0 + t(\mathbf{x} - \mathbf{x}_0))^\top (\mathbf{x} - \mathbf{x}_0) dt$$

Modeling $\mathbf{s}(\mathbf{x})$ is as expressive as modeling $p(\mathbf{x})$ itself, while often more tractable for generative modeling.

Outline

01 Langevin Dynamics

02 **Score Matching**

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Denoising Score Matching

Let us perturb the original data $\mathbf{x} \sim p_{\text{data}}(\mathbf{x})$ with Gaussian noise:

$$\mathbf{x}_\sigma = \mathbf{x} + \sigma \cdot \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I}), \quad q(\mathbf{x}_\sigma | \mathbf{x}) = \mathcal{N}(\mathbf{x}, \sigma^2 \cdot \mathbf{I})$$

$$q(\mathbf{x}_\sigma) = \int q(\mathbf{x}_\sigma | \mathbf{x}) p_{\text{data}}(\mathbf{x}) d\mathbf{x}.$$

Assumption

The solution to

$$\frac{1}{2} \mathbb{E}_{q(\mathbf{x}_\sigma)} \left\| \mathbf{s}_{\boldsymbol{\theta}, \sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

satisfies $\mathbf{s}_{\boldsymbol{\theta}, \sigma}(\mathbf{x}_\sigma) \approx \mathbf{s}_{\boldsymbol{\theta}, 0}(\mathbf{x}_0) = \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x})$ if σ is sufficiently small.

- For small σ, the noisy-data score $\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma)$ nearly matches the original-data score $\nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x})$.
- The score function $\mathbf{s}_{\boldsymbol{\theta}, \sigma}(\mathbf{x}_\sigma)$ is parameterized by σ.
- **Note:** We don't know $q(\mathbf{x}_\sigma)$, just as we don't know $p_{\text{data}}(\mathbf{x})$.

Denoising Score Matching

We cannot evaluate the marginal score $\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma)$. The theorem lets us replace it in the training objective with the conditional score $\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x})$.

Theorem

Under mild regularity conditions, the last term is independent of $\boldsymbol{\theta}$:

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) \right\|_2^2$$
$$= \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}) \right\|_2^2 + \text{const}$$

Gradient of the Noise Kernel

$$\mathbf{x}_\sigma = \mathbf{x} + \sigma \cdot \boldsymbol{\epsilon}, \qquad q(\mathbf{x}_\sigma|\mathbf{x}) = \mathcal{N}(\mathbf{x}, \sigma^2 \cdot \mathbf{I})$$
$$\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}) = -\frac{\mathbf{x}_\sigma - \mathbf{x}}{\sigma^2} = -\frac{\boldsymbol{\epsilon}}{\sigma}$$

Same minimizers, computable target: train $\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma)$ against $-\boldsymbol{\epsilon}/\sigma$, which we know because we added the noise ourselves.

Denoising Score Matching: Before the Proof

1. Start: Learn the Data Score

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \left\| \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}) - \nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x}) \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

2. Perturb the Data: Learn the Smoothed Score

Target: the unknown score of the noisy-data marginal $q(\mathbf{x}_\sigma)$.

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

3. Apply the Theorem: Same Minimizers, Known Target

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}) \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

Sample $\mathbf{x}_\sigma \sim q(\mathbf{x}_\sigma|\mathbf{x})$ via $\mathbf{x}_\sigma = \mathbf{x} + \sigma\boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$:

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{\boldsymbol{\epsilon}\sim\mathcal{N}(0,\mathbf{I})} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x} + \sigma \cdot \boldsymbol{\epsilon}) - \left(-\frac{\boldsymbol{\epsilon}}{\sigma} \right) \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

After Training: Sample with the Learned Score

$$\mathbf{x}_{l+1} = \mathbf{x}_l + \frac{\eta}{2} \cdot \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_l) + \sqrt{\eta} \cdot \boldsymbol{\epsilon}_l, \qquad \boldsymbol{\epsilon}_l \sim \mathcal{N}(0, \mathbf{I}).$$

Denoising Score Matching

Theorem

Under mild regularity conditions, the last term is independent of $\boldsymbol{\theta}$:

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) \right\|_2^2 = \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}) \right\|_2^2 + \text{const}.$$

Proof

For any integrable h, marginalize the joint distribution:

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} h(\mathbf{x}_\sigma) = \int \left(\int q(\mathbf{x}_\sigma|\mathbf{x}) p_{\text{data}}(\mathbf{x}) d\mathbf{x} \right) h(\mathbf{x}_\sigma) d\mathbf{x}_\sigma = \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} h(\mathbf{x}_\sigma).$$

Expand the squared error:

$$\begin{aligned} &\mathbb{E}_{q(\mathbf{x}_\sigma)} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) \right\|_2^2 \\ = \mathbb{E}_{q(\mathbf{x}_\sigma)} &\left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) \right\|_2^2 + \underbrace{\mathbb{E}_{q(\mathbf{x}_\sigma)} \left\| \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) \right\|_2^2}_{C_1} - 2\mathbb{E}_{q(\mathbf{x}_\sigma)} \left[\mathbf{s}_{\boldsymbol{\theta},\sigma}^\top(\mathbf{x}_\sigma) \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) \right]. \end{aligned}$$

The score-norm term C_1 is independent of $\boldsymbol{\theta}$.

Next: rewrite the last cross term $-2\mathbb{E}_{q(\mathbf{x}_\sigma)} \left[\mathbf{s}_{\boldsymbol{\theta},\sigma}^\top(\mathbf{x}_\sigma) \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) \right]$ using the conditional score.

Denoising Score Matching

Theorem

Under mild regularity conditions, the last term is independent of $\boldsymbol{\theta}$:

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) \right\|_2^2 = \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}) \right\|_2^2 + \text{const}.$$

Proof (Continued)

Use marginalization and $\nabla q = q \nabla \log q$ to rewrite the cross term:

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \left[\mathbf{s}_{\boldsymbol{\theta},\sigma}^\top(\mathbf{x}_\sigma) \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) \right] = \int \mathbf{s}_{\boldsymbol{\theta},\sigma}^\top(\mathbf{x}_\sigma) \nabla_{\mathbf{x}_\sigma} q(\mathbf{x}_\sigma) d\mathbf{x}_\sigma$$

$$= \int \mathbf{s}_{\boldsymbol{\theta},\sigma}^\top(\mathbf{x}_\sigma) \nabla_{\mathbf{x}_\sigma} \left(\int q(\mathbf{x}_\sigma|\mathbf{x}) p_{\text{data}}(\mathbf{x}) d\mathbf{x} \right) d\mathbf{x}_\sigma = \iint p_{\text{data}}(\mathbf{x}) \mathbf{s}_{\boldsymbol{\theta},\sigma}^\top(\mathbf{x}_\sigma) \nabla_{\mathbf{x}_\sigma} q(\mathbf{x}_\sigma|\mathbf{x}) \, d\mathbf{x} \, d\mathbf{x}_\sigma$$

$$= \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \left[\mathbf{s}_{\boldsymbol{\theta},\sigma}^\top(\mathbf{x}_\sigma) \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}) \right].$$

Substitute and complete the square:

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) \right\|_2^2 = \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \left[\left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) \right\|_2^2 - 2 \mathbf{s}_{\boldsymbol{\theta},\sigma}^\top(\mathbf{x}_\sigma) \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}) \right] + C_1$$

$$= \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}) \right\|_2^2 + \text{const}.$$

What Does Denoising Score Matching Learn?

Why the theorem works: MSE learns the average conditional target.

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})}\mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})}\left\|\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma}\log q(\mathbf{x}_\sigma|\mathbf{x})\right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

The model sees only $\mathbf{x}_\sigma$; the target also depends on $\mathbf{x}$.
For each $\mathbf{x}_\sigma$, the MSE-optimal prediction $\mathbf{s}^*_\sigma(\mathbf{x}_\sigma)$ is the average target:

$$\mathbf{s}^*_\sigma(\mathbf{x}_\sigma) = \mathbb{E}_{q(\mathbf{x}|\mathbf{x}_\sigma)}\left[\nabla_{\mathbf{x}_\sigma}\log q(\mathbf{x}_\sigma|\mathbf{x})\right]$$

From the Optimal Prediction to the Marginal Score

$$\begin{aligned}
\mathbf{s}^*_\sigma(\mathbf{x}_\sigma) &= \int \nabla_{\mathbf{x}_\sigma}\log q(\mathbf{x}_\sigma|\mathbf{x}) \cdot \frac{q(\mathbf{x}_\sigma|\mathbf{x})p_{\text{data}}(\mathbf{x})}{q(\mathbf{x}_\sigma)}d\mathbf{x} \\
&= \frac{1}{q(\mathbf{x}_\sigma)}\int \nabla_{\mathbf{x}_\sigma} q(\mathbf{x}_\sigma|\mathbf{x})p_{\text{data}}(\mathbf{x})d\mathbf{x} \\
&= \frac{\nabla_{\mathbf{x}_\sigma}\int q(\mathbf{x}_\sigma|\mathbf{x})p_{\text{data}}(\mathbf{x})d\mathbf{x}}{q(\mathbf{x}_\sigma)} = \nabla_{\mathbf{x}_\sigma}\log q(\mathbf{x}_\sigma).
\end{aligned}$$

Different conditional targets, one marginal score: $\mathbf{s}^*_\sigma(\mathbf{x}_\sigma) = \nabla_{\mathbf{x}_\sigma}\log q(\mathbf{x}_\sigma)$.
Variation around this average adds only a model-independent constant to the loss.

Tweedie's Formula: From Score to Predictions

For $\mathbf{x}_\sigma = \mathbf{x} + \sigma\boldsymbol{\epsilon}$, with independent $\mathbf{x} \sim p_{\text{data}}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$ and $\sigma > 0$, the posterior-average identity gives:

$$\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) = \mathbb{E}_{q(\mathbf{x}|\mathbf{x}_\sigma)}\left[\frac{\mathbf{x} - \mathbf{x}_\sigma}{\sigma^2}\right] = \frac{\mathbb{E}_{q(\mathbf{x}|\mathbf{x}_\sigma)}[\mathbf{x}] - \mathbf{x}_\sigma}{\sigma^2}.$$

Tweedie's Formula: Predict Clean Data

$$\mathbb{E}_{q(\mathbf{x}|\mathbf{x}_\sigma)}[\mathbf{x}] = \mathbf{x}_\sigma + \sigma^2 \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma).$$

Predict the Added Noise

$$\mathbb{E}[\boldsymbol{\epsilon} \mid \mathbf{x}_\sigma] = -\sigma \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma).$$

These are MSE-optimal conditional means, not necessarily the original $\mathbf{x}$ and $\boldsymbol{\epsilon}$.

Use the Trained Score Network

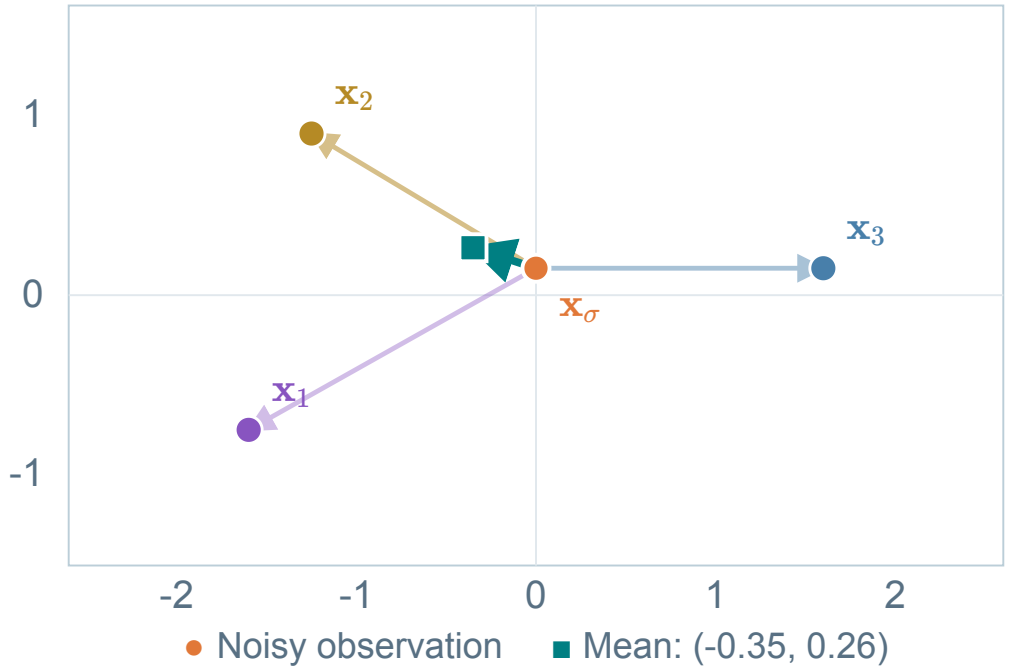
$$\hat{\mathbf{x}} = \mathbf{x}_\sigma + \sigma^2 \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma), \qquad \hat{\boldsymbol{\epsilon}} = -\sigma \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma).$$

One score model predicts both clean data and noise, approximating their conditional means.

From Conditional Scores to Denoising

Toy data: three equally likely clean points $\mathbf{x}_i$, with $\mathbf{x}_\sigma = \mathbf{x} + \sigma\boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$ and $\sigma = 1$.

Fixed noisy observation between the clean-data modes.



Conditional scores

$$\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}_i) = (\mathbf{x}_i - \mathbf{x}_\sigma)/\sigma^2$$

Bayes weights (priors 1/3 cancel)

$$w_i = q(\mathbf{x}_i|\mathbf{x}_\sigma) = \frac{\exp(-\|\mathbf{x}_\sigma - \mathbf{x}_i\|^2/(2\sigma^2))}{\sum_j \exp(-\|\mathbf{x}_\sigma - \mathbf{x}_j\|^2/(2\sigma^2))}$$

$q(\mathbf{x}_1\|\mathbf{x}_\sigma)$	$q(\mathbf{x}_2\|\mathbf{x}_\sigma)$	$q(\mathbf{x}_3\|\mathbf{x}_\sigma)$
22.9%	42.7%	34.4%

Marginal score = posterior-weighted average

$$\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) = \sum_i q(\mathbf{x}_i|\mathbf{x}_\sigma) \frac{\mathbf{x}_i - \mathbf{x}_\sigma}{\sigma^2}$$

Tweedie: from the score to the clean-data mean

$$\hat{\mathbf{x}} = \mathbf{x}_\sigma + \sigma^2 \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) = \sum_i w_i \mathbf{x}_i$$

Denoising averages plausible clean points: $\hat{\mathbf{x}} = \sum_i w_i \mathbf{x}_i$, possibly between modes.

Vincent P. A Connection Between Score Matching and Denoising Autoencoders, 2010
Efron B. Tweedie's Formula and Selection Bias, 2011

Denoising Score Matching: Method Overview

1. Start: Learn the Data Score

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \left\| \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}) - \nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x}) \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

2. Perturb the Data: Learn the Smoothed Score

Target: the unknown score of the noisy-data marginal $q(\mathbf{x}_\sigma)$.

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

3. Apply the Theorem: Same Minimizers, Known Target

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}) \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

Sample $\mathbf{x}_\sigma \sim q(\mathbf{x}_\sigma|\mathbf{x})$ via $\mathbf{x}_\sigma = \mathbf{x} + \sigma\boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$:

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{\boldsymbol{\epsilon}\sim\mathcal{N}(0,\mathbf{I})} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x} + \sigma \cdot \boldsymbol{\epsilon}) - \left(-\frac{\boldsymbol{\epsilon}}{\sigma} \right) \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

After Training: Sample with the Learned Score

$$\mathbf{x}_{l+1} = \mathbf{x}_l + \frac{\eta}{2} \cdot \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_l) + \sqrt{\eta} \cdot \boldsymbol{\epsilon}_l, \qquad \boldsymbol{\epsilon}_l \sim \mathcal{N}(0, \mathbf{I}).$$

Outline

01 Langevin Dynamics

02 Score Matching

Denoising Score Matching

Noise-Conditioned Score Network (NCSN)

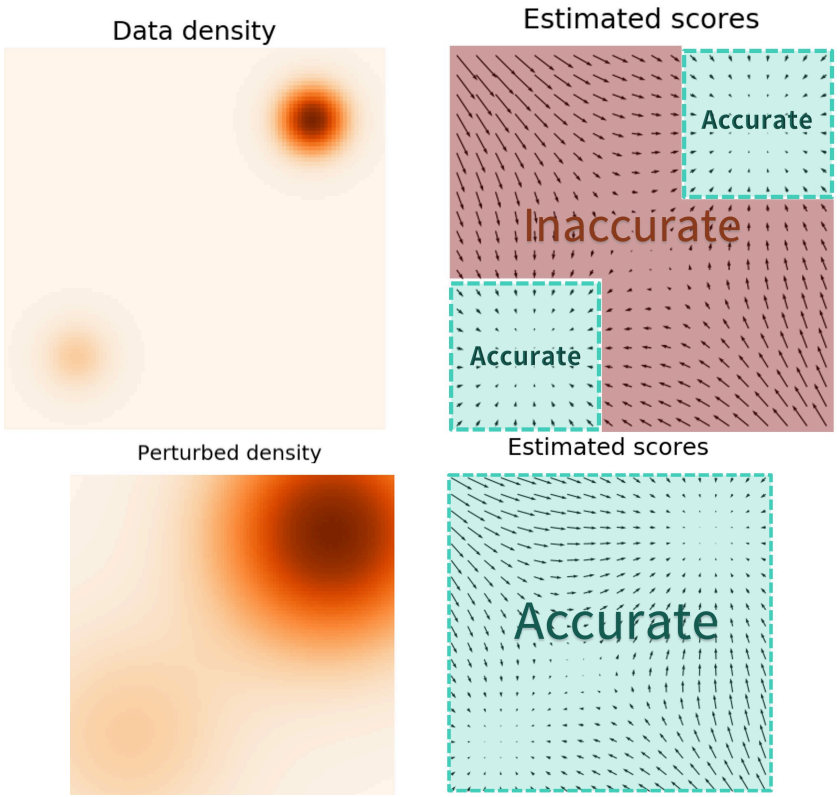
03 Forward Gaussian Diffusion Process

Denoising Score Matching

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})}\mathbb{E}_{\mathcal{N}(0,\mathbf{I})}\left\|\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}+\sigma\boldsymbol{\epsilon})+\frac{\boldsymbol{\epsilon}}{\sigma}\right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

$$\mathbf{x}_{l+1} = \mathbf{x}_l + \frac{\eta}{2}\cdot\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_l) + \sqrt{\eta}\cdot\boldsymbol{\epsilon}_l$$

- For **small** σ, $\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x})$ becomes inaccurate and Langevin dynamics fails to traverse modes.
- For **large** σ, robustness in low-density regions is achieved, but the model learns a distribution that is overly corrupted.

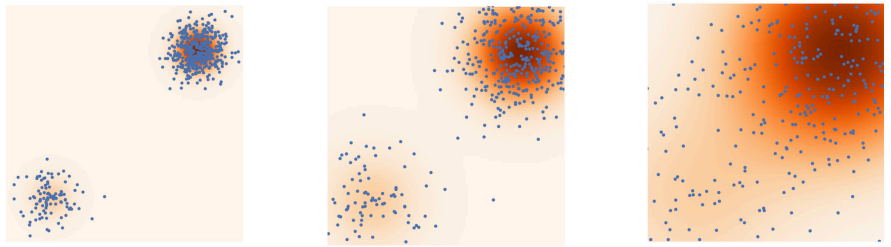


Noise-Conditioned Score Network (NCSN)

- Specify a sequence of noise levels: $\sigma_1 < \sigma_2 < \cdots < \sigma_T$.
- Perturb each data point with different noise levels: $\mathbf{x}_t = \mathbf{x} + \sigma_t \boldsymbol{\epsilon}$, so $\mathbf{x}_t \sim q(\mathbf{x}_t)$.
- Choose σ_1, σ_T such that:

$$q(\mathbf{x}_1) \approx p_{\text{data}}(\mathbf{x}), \qquad q(\mathbf{x}_T) \approx \mathcal{N}(0, \sigma_T^2 \mathbf{I})$$

$$\sigma_1 < \sigma_2 < \sigma_3$$





Noise-Conditioned Score Network (NCSN)

Train the denoising score function $\mathbf{s}_{\boldsymbol{\theta},\sigma_t}(\mathbf{x}_t)$ for each noise level σ_t using a unified weighted objective:

$$\sum_{t=1}^{T} \sigma_t^2 \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x})} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma_t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t|\mathbf{x}) \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

The target is $-\boldsymbol{\epsilon}/\sigma_t$. The weight σ_t^2 compensates its scale:

$$\sigma_t^2 \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma_t}(\mathbf{x}_t) + \frac{\boldsymbol{\epsilon}}{\sigma_t} \right\|^2 = \left\| -\sigma_t \mathbf{s}_{\boldsymbol{\theta},\sigma_t}(\mathbf{x}_t) - \boldsymbol{\epsilon} \right\|^2.$$

Training

1. Sample $\mathbf{x}_0 \sim p_{\text{data}}(\mathbf{x})$, $t \sim U\{1, T\}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$.
2. Compute noisy image $\mathbf{x}_t = \mathbf{x}_0 + \sigma_t \boldsymbol{\epsilon}$.
3. Compute loss $\mathcal{L} = \sigma_t^2 \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma_t}(\mathbf{x}_t) + \frac{\boldsymbol{\epsilon}}{\sigma_t} \right\|^2$.
4. Update $\boldsymbol{\theta}$ via stochastic gradient descent.

How do we sample from such a model?

Noise-Conditioned Score Network (NCSN)

Inner loop (l): run Langevin dynamics at fixed σ_t, targeting $q(\mathbf{x}_t)$.
Outer loop ($t = T, \dots, 1$): lower the noise and reuse samples at the next level.

Sampling (Annealed Langevin Dynamics)

1. Sample $\mathbf{x}_T^0 \sim \mathcal{N}(0, \sigma_T^2 \mathbf{I}) \approx q(\mathbf{x}_T)$.
2. Update $\mathbf{x}_t^l$ via L steps of Langevin dynamics at each noise level σ_t:

$$\mathbf{x}_t^l = \mathbf{x}_t^{l-1} + \frac{\eta_t}{2} \mathbf{s}_{\boldsymbol{\theta}, \sigma_t}(\mathbf{x}_t^{l-1}) + \sqrt{\eta_t} \boldsymbol{\epsilon}_t^l.$$

3. Update $\mathbf{x}_{t-1}^0 := \mathbf{x}_t^L$ and proceed to σ_{t-1} if $t > 1$.

Why Gradually?

High noise helps explore different modes; lower noise restores detail. Nearby noise levels let us reuse samples instead of restarting.



Outline

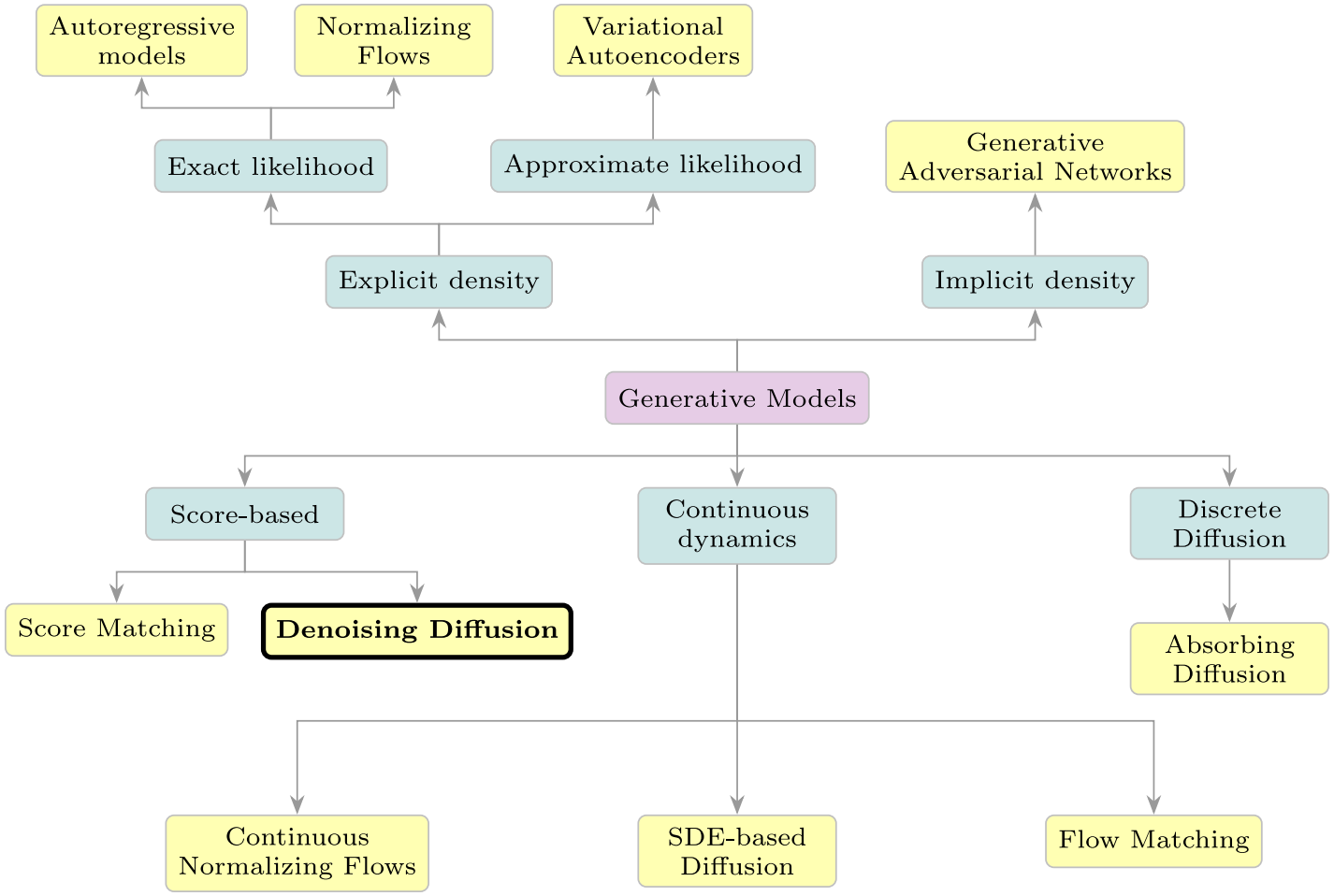
01 Langevin Dynamics

02 Score Matching

Denoising Score Matching
Noise-Conditioned Score Network (NCSN)

03 Forward Gaussian Diffusion Process

Generative Models Taxonomy



Forward Gaussian Diffusion Process

Let $\mathbf{x}_0 = \mathbf{x} \sim p_{\text{data}}(\mathbf{x})$, $\beta_t \ll 1$. Define a Markov chain:

$$\mathbf{x}_t = \sqrt{1-\beta_t}\mathbf{x}_{t-1} + \sqrt{\beta_t}\boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(0, \mathbf{I})$$

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I})$$

Langevin Dynamics

$$\mathbf{x}_{l+1} = \mathbf{x}_l + \frac{\eta}{2} \cdot \nabla_{\mathbf{x}_l} \log p_{\boldsymbol{\theta}}(\mathbf{x}_l) + \sqrt{\eta}\boldsymbol{\epsilon}_l, \quad \boldsymbol{\epsilon}_l \sim \mathcal{N}(0, \mathbf{I})$$

$$\begin{aligned}
\mathbf{x}_t &= \sqrt{1-\beta_t}\,\mathbf{x}_{t-1} + \sqrt{\beta_t}\boldsymbol{\epsilon}_t \\
&\approx \left(1 - \frac{\beta_t}{2}\right)\mathbf{x}_{t-1} + \sqrt{\beta_t}\boldsymbol{\epsilon}_t \\
&= \mathbf{x}_{t-1} + \frac{\beta_t}{2}(-\mathbf{x}_{t-1}) + \sqrt{\beta_t}\boldsymbol{\epsilon}_t
\end{aligned}$$

From a Markov chain to Langevin dynamics: for small β_t, this Gaussian chain approximates LD targeting $\mathcal{N}(0, \mathbf{I})$, with step size $\eta = \beta_t$ and Gaussian score $-\mathbf{x}_{t-1}$.

Forward Diffusion: Sample at Any Timestep

Fix a schedule $0 < \beta_t < 1$. The noises $\boldsymbol{\epsilon}_t \sim \mathcal{N}(0, \mathbf{I})$ are independent across t and of $\mathbf{x}_0$.

$$\mathbf{x}_t = \sqrt{1-\beta_t}\mathbf{x}_{t-1} + \sqrt{\beta_t}\boldsymbol{\epsilon}_t.$$

Exact Conditional Distribution

Let $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$. Then

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t}\,\mathbf{x}_0, (1-\bar{\alpha}_t)\mathbf{I}).$$

Sample $\mathbf{x}_t$ directly from $\mathbf{x}_0$, without simulating the intermediate steps:

$$\begin{aligned}
\mathbf{x}_t &= \sqrt{\alpha_t}\mathbf{x}_{t-1} + \sqrt{1-\alpha_t}\boldsymbol{\epsilon}_t \\
&= \sqrt{\alpha_t\alpha_{t-1}}\mathbf{x}_{t-2} + (\sqrt{\alpha_t(1-\alpha_{t-1})}\boldsymbol{\epsilon}_{t-1} + \sqrt{1-\alpha_t}\boldsymbol{\epsilon}_t) \\
&= \sqrt{\alpha_t\alpha_{t-1}}\mathbf{x}_{t-2} + \sqrt{1-\alpha_t\alpha_{t-1}}\boldsymbol{\epsilon}'_t \\
&= \ldots = \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I}).
\end{aligned}$$

When Does Only Gaussian Noise Remain?

The closed-form conditional already tells us how the signal disappears:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I}), \quad \boldsymbol{\epsilon} \perp \mathbf{x}_0.$$

The Schedule Must Remove the Signal

For $0 < \beta_t < 1$, the required condition is

$$\bar{\alpha}_t = \prod_{s=1}^{t}(1-\beta_s) \longrightarrow 0 \quad \Longleftrightarrow \quad \sum_{t=1}^{\infty}\beta_t = \infty.$$

Many steps alone are not enough: $\beta_t = 2^{-t}$ leaves $\bar{\alpha}_\infty > 0$.

$$\sqrt{\bar{\alpha}_t} \to 0, \qquad \sqrt{1-\bar{\alpha}_t} \to 1 \quad \Longrightarrow \quad \mathbf{x}_t \xrightarrow{d} \mathcal{N}(0, \mathbf{I}).$$

The limit no longer depends on the initial data distribution p_{data}.

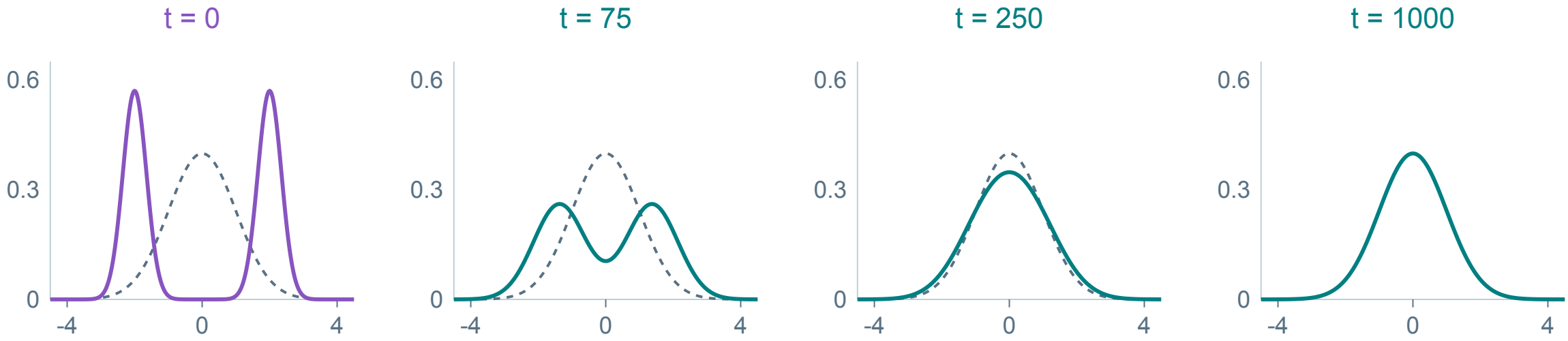
> **For finite T:** choose $\bar{\alpha}_T \ll 1$ with negligible residual signal $\sqrt{\bar{\alpha}_T}\mathbf{x}_0$.
> Then $q(\mathbf{x}_T) \approx \mathcal{N}(0, \mathbf{I})$: the terminal distribution is approximately standard Gaussian.

Forward Diffusion: From Data to Noise

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}\left(\sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I}\right);$$

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}\left(\sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1-\bar{\alpha}_t)\mathbf{I}\right)$$

$$p_{\text{data}}(x_0) = \tfrac{1}{2}\mathcal{N}(-2, 0.35^2) + \tfrac{1}{2}\mathcal{N}(2, 0.35^2), \qquad \beta_t = 0.01$$

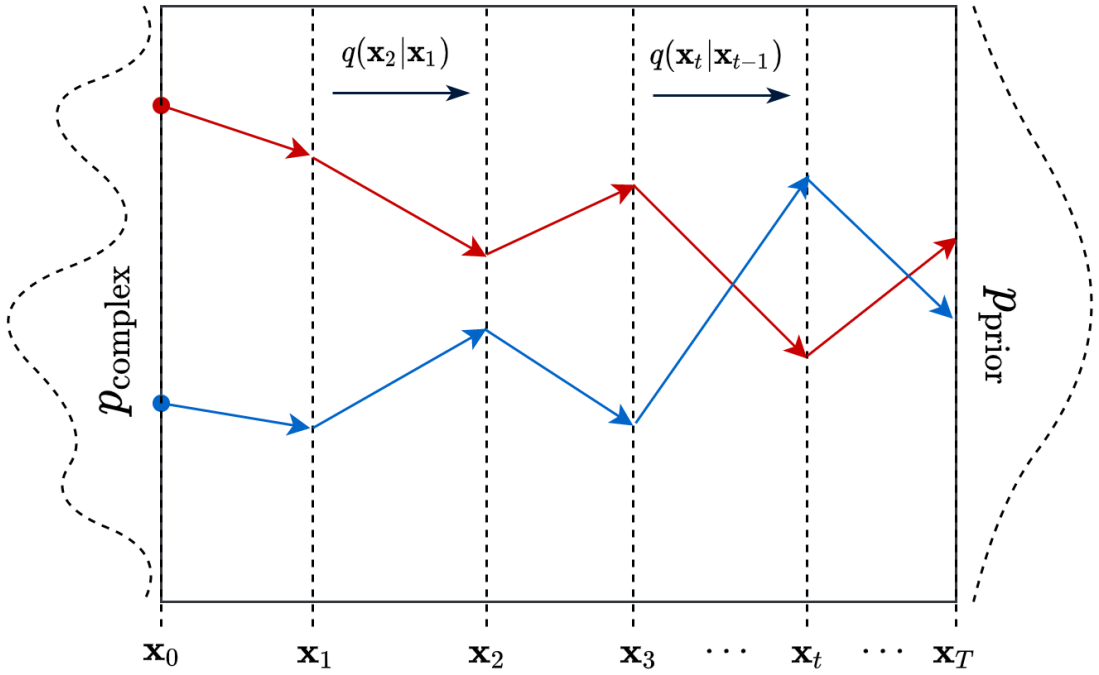


---- Standard Gaussian reference $\mathcal{N}(0,1)$.

The plots show marginal densities; the kernels above are conditional.

Forward Gaussian Diffusion Process

Diffusion describes the migration of particles from regions of high density to those of low density.



1. $\mathbf{x}_0 = \mathbf{x} \sim p_{\text{data}}(\mathbf{x})$
2. $\mathbf{x}_t = \sqrt{1-\beta_t}\mathbf{x}_{t-1} + \sqrt{\beta_t}\boldsymbol{\epsilon}_t,$
   $\boldsymbol{\epsilon}_t \sim \mathcal{N}(0, \mathbf{I}),\ t \geq 1$
3. Choose $\bar{\alpha}_T \ll 1$ to suppress $\sqrt{\bar{\alpha}_T}\mathbf{x}_0$.
   Then $q(\mathbf{x}_T) \approx \mathcal{N}(0, \mathbf{I})$.

If this process can be reversed, we can sample from $p_{\text{data}}(\mathbf{x})$ by starting from noise $\mathcal{N}(0, \mathbf{I}) \approx q(\mathbf{x}_T)$.
Our goal now becomes inverting this diffusion.

Denoising Score Matching

NCSN

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_0, \sigma_t^2\mathbf{I}), \quad q(\mathbf{x}_1) \approx p_{\text{data}}(\mathbf{x}), \quad q(\mathbf{x}_T) \approx \mathcal{N}(0, \sigma_T^2\mathbf{I})$$

$$\nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t|\mathbf{x}) = -\frac{\mathbf{x}_t - \mathbf{x}}{\sigma_t^2}$$

Gaussian Diffusion

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1-\bar{\alpha}_t)\mathbf{I}), \quad q(\mathbf{x}_1) \approx p_{\text{data}}(\mathbf{x}), \quad q(\mathbf{x}_T) \approx \mathcal{N}(0, \mathbf{I})$$

$$\nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t|\mathbf{x}_0) = -\frac{\mathbf{x}_t - \sqrt{\bar{\alpha}_t}\mathbf{x}_0}{1-\bar{\alpha}_t}$$

Theorem (Denoising Score Matching)

$$\mathbb{E}_{q(\mathbf{x}_t)} \left\| \mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t) \right\|_2^2$$
$$= \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x})} \left\| \mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t|\mathbf{x}) \right\|_2^2 + \text{const}$$

Note: Annealed Langevin dynamics applies to diffusion, too.
NCSN with annealed Langevin dynamics is an early score-based generative framework (2019–2020).

Summary

- Langevin dynamics combines log-density gradients and Gaussian noise to sample approximately from an unnormalized model.
- Score matching minimizes Fisher divergence to estimate the score function.
- Denoising score matching makes the score objective estimable using corrupted samples and the known corruption kernel.
- Tweedie's formula converts the score into MSE-optimal clean-data and noise predictions.
- Noise-Conditioned Score Networks balance score losses with weights σ_t^2 and sample with annealed Langevin dynamics.
- The forward Gaussian diffusion process admits a closed-form conditional distribution for any timestep.
- With a suitable noise schedule, the forward process converges to standard Gaussian noise; denoising score matching also applies to these diffusion marginals.